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Machine Learning-Based Diagnosis and Detection of Liver Cancer: An Approach Enhancement

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Abstract --- One of the most important organs in our body is the liver, yet liver cancer incidence has been sharply rising in recent years, primarily as a result of poor lifestyles. In order to accurately identify the ailment, this study suggests a diagnosis system based on multiple machine-learning techniques. After managing the dataset's missing values, data encoding, and segregation are carried out for pre-processing. Upon conducting feature scaling on the used dataset, machine learning models are applied. Out of the seven models utilized, the findings demonstrate that SVM and LGB classifiers perform exceptionally well in terms of the highest accuracy and least amount of loss, while the Random Forest approach surpassed all others when recall, F1 score, and precision are to be considered.

Keywords --- Liver Cancer, Machine Learning, SVM, Random Forest

I. INTRODUCTION

Numerous organs in the human body are necessary for survival. The liver is one of these organs. By creating bile, the liver plays a key part in helping the body get rid of harmful chemicals and aids in food digestion [1]. It is crucial for the synthesis of proteins in hematology. Its size is approximately the same as a football, and it also serves as a storehouse for iron and copper. Liver cancer is the fourth most frequent cause of mortality in those with invasive cancer [2].

Even though there are no early signs of liver cancer, unintentional weight loss, a decrease in appetite, nausea, jaundice, and a variety of other symptoms should be taken seriously [3]. By altering the patient's diets and boosting their immune systems, the condition may be prevented and treated. The disease is treated with immunotherapy, chemotherapy, and nanotechnology-based methods [4].

Primary liver cancer, which is further divided into two subgroups, is cancer that develops in the liver tissue. The most prevalent type is hepatocellular carcinoma (HCC), which develops when cancer cells proliferate in the tissues of the liver. Bile duct cancer, also known as cholangiocarcinoma, develops when the liver's bile ducts drain into the small intestine. Metastatic, which develops when the infected cells from other body parts move to the liver, is another name for secondary liver cancer [5-6].

Because most liver cancer patients receive a very late diagnosis, treatment can be challenging. Computed Tomography (CT) scans are popularly employed in detection. Early diagnosis is essential for providing patients with appropriate care in a timely manner [7-8]. Artificial Intelligence is the most recent emerging and useful technology and the subset of it is machine learning is a popular technique that teaches computers to act and make decisions like people. Using a variety of algorithms that have been trained to do so, this technology can identify liver cancer [9-12].

The purpose of the research work is to forecast the results of machine learning-based models for the early prediction and classification of liver cancer. There are four main components to the essay. An overview of the literature that has been done on the subject is covered in the first part. A description of the dataset used for training the models comes after this section. The proposed approach has also been addressed, and the outcomes are discussed in the paper's results section.

II. LITERATURE SURVEY

The research on the diagnosis and classification of liver cancer by various writers is discussed in the section that follows. This section analyses investigations of existing works and discusses newly proposed machine-learning algorithms for disease detection.

According to Afrin et al., algorithms like decision trees, random forests, KNN, logistic regression, AdaBoost, and SVM [13-14] were employed for the diagnosis of liver cancer [15]. The least absolute shrinkage and selection operator (LASSO) was used to choose the dataset's most valuable associated characteristics. With 94.2% accuracy, 92% precision, a 96% f1-score value, and 99% specificity, the decision tree method topped all others.

Ali et al. [16] designed a system that performs classification using SVM through Computer-Aided Diagnosis (CAD) while feature extraction is carried out by employing DWT. In order to improve the quality of the photos, pre-processing is first necessary. Wavelet feature extraction is done after performing liver segmentation using masking and a Gaussian filter. The SVM classifier is used to implement the HCC classification process

Similarly, an ML-based clinical decision support system (CDSS) was created by Choi et al. [17] and predicts the overall survival (OS) rate and early treatment options for the

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identification of a liver illness. Twenty pretreatment variables were picked from hospital records that contained data on pretreatment variables, initial therapy, and survival status. For treatment decision-making, the model used a six-step hierarchical classifier and achieved an accuracy of 81%.

An automated method for this disease's diagnosis was suggested by a different study by Das et al. [18]. In India, information on 123 patients was gathered from the IMS and SUM hospitals. The separation of the liver from the whole image is done by adaptive thresholding and the segmentation of cancerous lesions was done next. The type of liver cancer was classified from the collected features using MLP and C4.5 decision tree. The MLP classifier's accuracy was 89%, and the C4.5 classifier's accuracy was 95%.

Jasti et al. [19] proposed an image- processing Machine Learning based method that accurately diagnoses liver cancer in another article. First, the image noise is leveled using the histogram equalization technique. The region of interest is segmented using the k-means algorithm. Finally, the classification is carried out utilizing the ANN model, RBF-SVM, and Random Forest approach. The RBF-

SVM model was found to perform better than the other two.

An approach to identify the disease using ML techniques was put out by Kim et al. [20]. This technique searches for relevant markers in sensor arrays with extreme sensitivity, and a neural network and fuzzy neural network are utilized to detect the disease. A total of 395 samples were evaluated, and using NN and fuzzy NN, respectively, 99.19% and 98.19% of them were correctly identified.

Another ML technique to identify HCC was also proposed by Ksiazek et al. [21] and tested on data from 165 patients with 49 features. The first thing to perform is normalization, which entails: a) filling in missing values with the modal value of the given feature; b) filling in the average value for numerical attributes; and c) scaling numerical and categorical characteristics to a range from 0 to 1. Second, classification is carried out in three steps, with the cross-validation process involving feature selection and Genetic Algorithm parameter optimization. Out of the ten algorithms that were implemented, SVM produced the highest accuracy.

Singal et al. [22] also designed a methodology for diagnosing HCC using ML, which made use of data from 442 cases. Models that might forecast the development of HCC were created using regression analysis and ML techniques. Regression analysis underperformed the ML algorithm in terms of accuracy.

A technique to locate and separate liver cancers in 3D CT scan pictures was put forth by Huang et al. [23]. As a voxel classifier, a strong ELM algorithm is trained. A voxel is a representation of a substantial feature vector. A semi-automatic method

is used to segment the region of interest by randomly gathering samples in 3D space. The strategy produced effective segmentation and detection results.

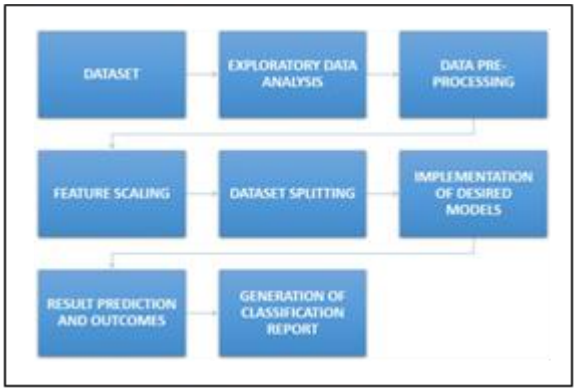

Fig. 1. Proposed Methodology

III. Dataset Used

Using data on individuals with liver illnesses, machine learning models were tested for effectiveness and their ability to forecast outcomes. The dataset for the proposed research project can be acquired from Kaggle. Along with some attribute data, the dataset consists of 20,000 training data points and 1,000 test data points. The attributes are age, gender, total bilirubin, direct bilirubin, alkaline phosphatase, alanine aminotransferase, aspartate aminotransferase, total proteins, albumin, albumin & globulin ratio.

IV. Proposed Methodology

We conducted research on the dataset using multiple machine-learning models after doing a literature review and extensive analytical examination. The author's method, the system's workflow, and the many methodologies employed in its design are all addressed in the following section. Fig. 1 shows the proposed methodology.

A. Exploratory Data Analysis

Firstly, the pandas' library is used to read the dataset. Then, using data visualization approaches, exploratory data analysis (EDA) graphically represents the data, analyses, and extracts its major properties. The research findings have been proposed based on these characteristics. Fig. 2 depicts the heatmap for liver cancer detection parameters and Fig. 3 represents the correlation between these parameters.

B. Data Pre-processing

It is done after data visualization in order to get the data ready for the model's training and testing. To determine how many columns have missing data and determine whether those columns are necessary or not, we utilize a simpleimputer class from the sklearn package. To use the autoencoder class to transform strings into numerics, the next step is to determine how many columns have non-numeric values. The data are finally normalized by data normalization

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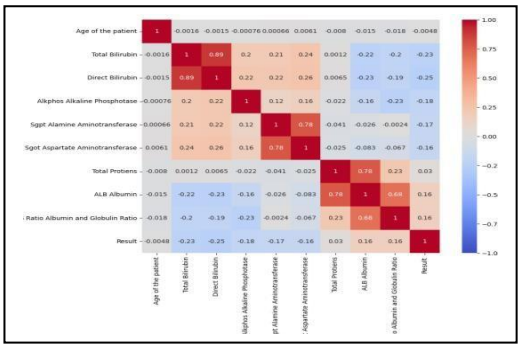

Fig. 2. Heatmap for Liver Cancer detection parameters

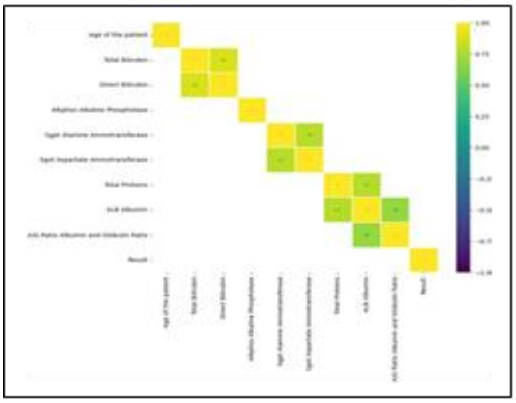

Fig. 3. Correlation for Liver Cancer detection parameters

C. Splitting Database

Next, a training and a test set are created from the dataset. It is divided employing a 75/25 ratio.

D. Feature Scaling

Feature scaling can be applied to deal with out- of-range data in order to obtain correct results. By utilizing the MinMaxScaler() class with the default parameters, MinMaxScaler is initially used to normalize the input variables. Additionally, the input variables are standardized using the StandardScaler Transform. PCA can also be used as required.

TABLE I. PERFORMANCE ANALYSIS USING ACCURACY AND LOSS

Models	**Liver Cancer Detection**	
	Accuracy	***Loss***
KNN	0.91	0.094
Naïve Bayes	0.66	0.34
Random Forest	1.00	0.0029
Logistic regression	0.98	0.019
SVM	1.00	0.0
CatBoost	0.98	0.016
LGB Classifier	1.00	0.0

E. Model Implementation and Prediction

KNN, Naive Bayes classifier, Random Forest, logistic regression, SVM, CatBoost, and LGB classifier, have been applied to the provided dataset. On the basis of the provided dataset of independent factors, all of these models calculated the likelihood that liver cancer exists. As indicated in the next section, many parameters have been utilized to produce model predictions and results.

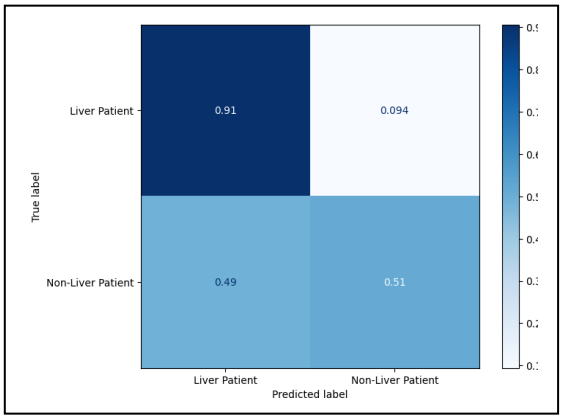

Fig. 3. KNN model

V. RESULTS AND OUTCOMES

The proposed study used a variety of parameters to analyze model prediction. As indicated in the tables, two separate sets of parameters are used to represent performance analysis. The final results are depicted using a confusion matrix.

TABLE II. PERFORMANCE ANALYSIS USING PRECISION, RECALL, AND F1-SCORE

Models	**Liver Cancer Detection**		
	Precision	***Recall***	***F1-score***
KNN	0.82	0.91	0.86
Naïve Bayes	0.95	0.34	0.51
Random Forest	1.00	1.00	1.00
Logistic regression	0.72	0.98	0.83
SVM	0.72	1.00	0.83
CatBoost	0.92	0.98	0.95
LGB Classifier	0.72	1.00	0.83

A. Accuracy and Loss

Loss is a parameter that shows the total number of errors in the model, whereas accuracy evaluates the prediction efficacy of the model and is used to determine the level of success or failure that the model is doing.

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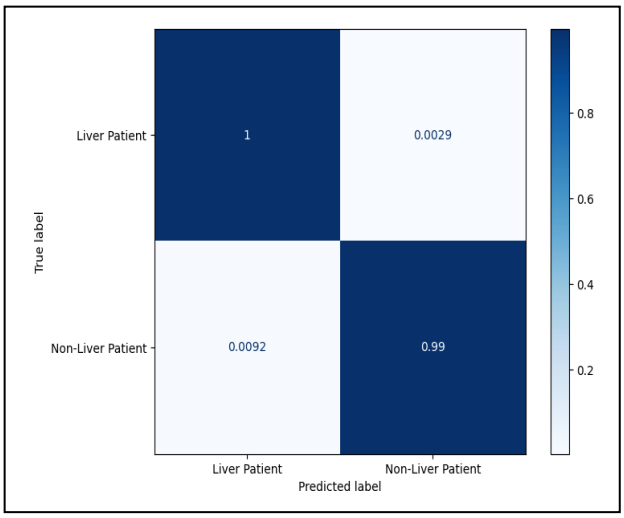

Fig. 4. Random Forest classifier

A high loss value denotes an ineffective model. The model performs best when it exhibits high accuracy and minimal loss; in this scenario, the model only produces tiny mistakes in a small subset of the data. SVM and LGB classifiers perform rather well in accurately predicting the outcomes with no loss value, whereas the Naive Bayes classifier performs poorly with maximum loss and minimum accuracy value on the provided dataset, as shown in Table I.

B. *Precision, Recall, and F1-Score*

While recall, also known as sensitivity, is the ratio of accurately anticipated positive outcomes to all positive cases in the data, precision is a measure of the number of positive predictions made that are accurate. Both the precision and recall levels must be high for the model to be effective. F1 score is another metric that is the mean of previously stated two parameters. Table II shows that the Random Forest classifier outperforms all other algorithms on the provided dataset with maximum values of all three parameters.

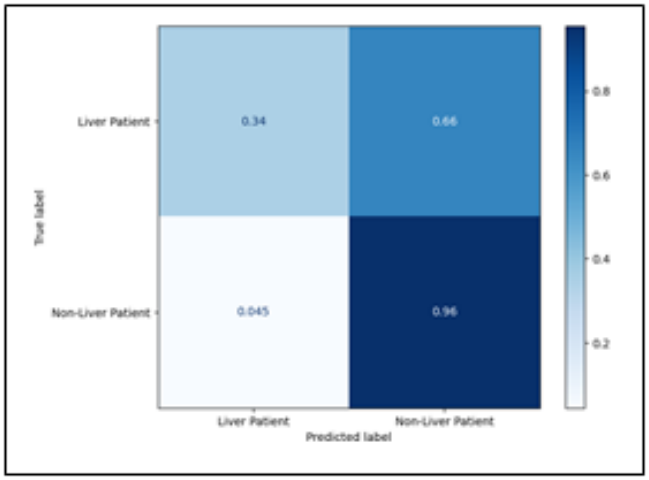

Fig. 5. Naïve Bayes model

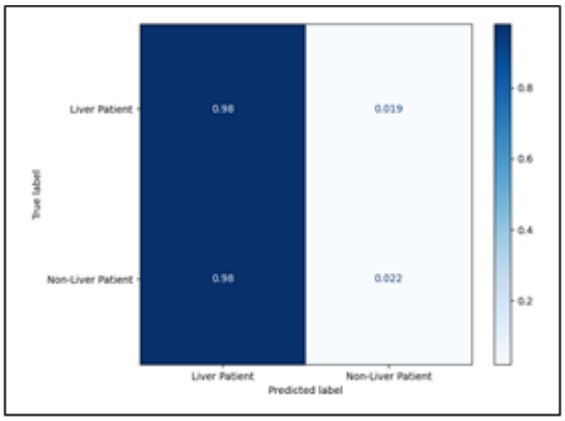

Fig. 6. Logistic Regression Model

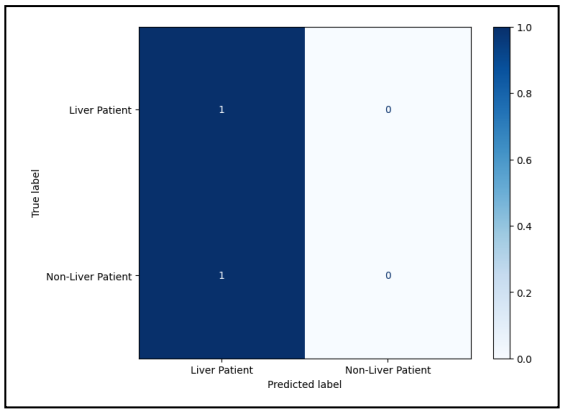

Fig. 7. SVM model

C. *Confusion Matrix Representation*

An illustration of classifier performance based on the four values of TP, FP, TN, and FN can be provided via a confusion matrix. These are set up against one another to show how they relate.

1) KNN Model: K-Nearest Neighbour is an easy-to-use, non-parametric supervised machine learning technique that, assuming similarity between the new data and old instances, assigns the new case to the category that matches the existing ones most closely. Fig. 4 depicts the confusion matrix for the KNN model.

2) Naïve Bayes Classifier*:* In order to solve classification difficulties, the Naive Bayes method is a powerful classification system that generates speedy predictions as shown in Fig. 5. It uses the Bayes theorem and is a probabilistic classifier.

3) Random Forest Classifier*:* This well- known SL technique, which is based on ensemble learning, uses many decision trees on smaller datasets and averages the results to improve prediction accuracy shown in

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Fig. 6. More trees in a forest result in higher accuracy.

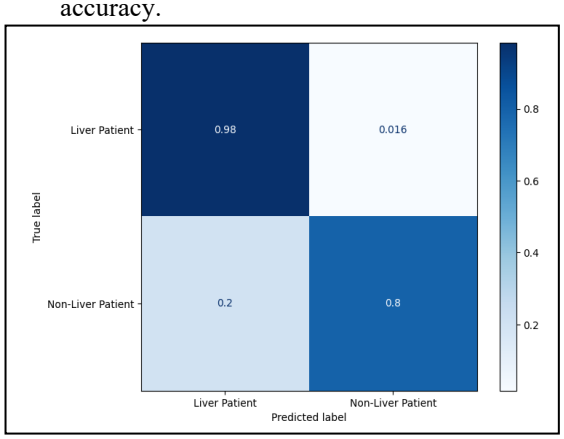

Fig. 8. CatBoost

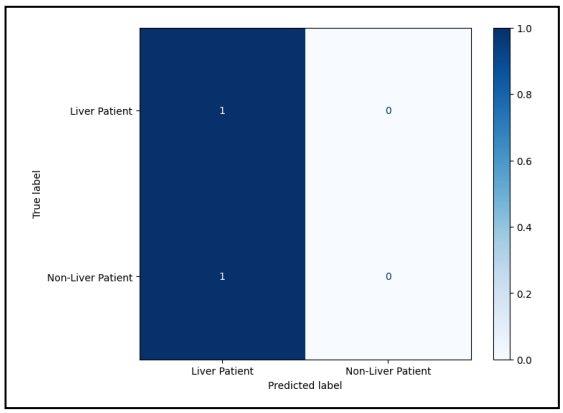

Fig. 9. LGB classifier

4) Logistic Regression*:* Logistic regression uses a set of independent features to forecast the categorical dependent feature. The output will be a category. It provides probabilistic values that range from 0 to 1. The confusion matrix for this model is shown in Fig. 7.

5) SVM Model: The support vector machine, which is mostly employed for classification issues, is another well-known SL algorithm. It establishes a decision boundary that can categorize a space with n dimensions, simplifying new instances as they emerge. It selects the extreme points or vectors that aid in the creation of the hyperplane based on statistical learning theory. The results are shown in Fig. 8.

6) CatBoost Algorithm: A more modern boosting technique that is extremely dependable, accurate, and quick is called the categorical boosting algorithm. It is created to operate with features connected to a category. The algorithm can handle both categorical and numerical information, performing label encoding on its own. According to Fig. 9, the algorithm performs effectively with the supplied dataset.

7) LGB Classifier: GOSS and EFB are two gradient-boosting algorithms built on decision trees that LightGBM uses to improve the model's efficiency. Due to its excellent speed and performance, it is used for classification, ranking, and other ML tasks. Fig. 10. Depicts the confusion matrix for the LGB classifier.

VI. Conclusion and Future Scope

Liver is among the several organs that make up the human body. The liver is responsible for both food digestion and the removal of toxins from our bodies. Similar to how many organs are affected by cancer, liver disease is also one of the deadliest and most common. It is so critical to identify it quickly and begin treatment. The accuracy of these systems relies on the model utilized; however, numerous AI and ML techniques have already been used to detect the disease successfully. To test which machine learning model performed best on the selected dataset, we proposed a system that included seven ML models and performance parameters utilized in a confusion matrix to depict the results for each model. It has been discovered that the Random Forest works best as all the parameters show the highest values. The future will hold considerable possibilities for changing liver cancer diagnosis and treatment because of ongoing research and technology improvements.

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